CEED 2027 Big Part A Workbook: 35-Questions-Per-Day System & Master Question Bank

350 Brand-New Original CEED-2027 Questions Across 10 Full 35-Question Timed Papers (6 NAT + 8 MSQ + 21 MCQ = 119 Marks/Day in 35–45 Mins), 88-Day Schedule, Custom Vector Diagrams & Complete Solutions

This master Part A workbook is engineered specifically for your daily **35-to-45-minute** preparation block from **October 21, 2026 through January 16, 2027 (88 Days)**. Because the official CEED 2024–2026 Part A examination requires solving 44 questions in 60 minutes (1.36 minutes/question), solving **35 questions in 35 to 45 minutes** (1.00 to 1.28 minutes/question) trains you at the exact cognitive velocity required in the examination hall while preserving the authentic 2024–2026 section ratio and partial-marking mechanics.

Blueprint of Your Daily 35-Question Part A Paper (119 Marks Total)

Part A Section	Official Exam (60m)	Your Daily Sprint (35–45m)	Marking Scheme (CEED 2024–2026 Standard)	Target Time Split
Section I: NAT (Numerical Answer Type)	8 Qs × 4m = 32m	**6 Qs × 4m = 24 Marks** (Q.01 to Q.06)	**+4 Marks** for exact numerical answer; **0 Negative** marking; 0 for unattempted.	**11–12 Minutes** (~1.9 mins/Q)
Section II: MSQ (Multiple Select Question)	10 Qs × 4m = 40m	**8 Qs × 4m = 32 Marks** (Q.07 to Q.14)	**+4** (all correct), **+3** (3 of 4), **+2** (2 of 3+), **+1** (1 of 2+); **−1 Negative** if any wrong option chosen.	**10–12 Minutes** (~1.4 mins/Q)
Section III: MCQ (Multiple Choice Question)	26 Qs × 3m = 78m	**21 Qs × 3m = 63 Marks** (Q.15 to Q.35)	**+3 Marks** for single correct option; **−0.5 Negative** for wrong option; 0 for unattempted.	**14–21 Minutes** (~0.8 mins/Q)
Daily Sprint Total	44 Qs = 150 Marks	**35 Qs = 119 Marks**	Target Qualifying Score: **65+ / 119** | Top-100 Rank Target: **88+ / 119**	**35–45 Minutes**

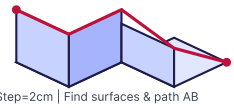
Section I: 10 Full-Length Original 35-Question Practice Papers (350 Questions)

Each of the ten full-length papers below contains **35 brand-new original CEED-style questions** (Q.01–Q.06 NAT, Q.07–Q.14 MSQ, Q.15–Q.35 MCQ) synthesized from the 10-year longitudinal patterns identified in CEED 2017–2026.

Practice Paper 01: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

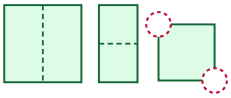
Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)

FIG P1.1: 3D SOLID & PATH (Q.01)



Step=2cm | Find surfaces & path AB

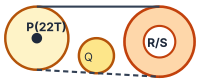
FIG P1.2: FOLD & CUT (Q.03, Q.11)



Sheet=9×9cm, Cut r=2cm

Unfolded cutouts & symmetry

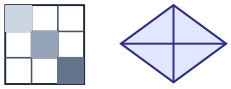
FIG P1.3: MECHANICS (Q.05, Q.13)



Load=2040N | P=4.0 turns

Mechanical advantage & gear ratio

FIG P1.4: VIEWS & MATRIX (Q.15–35)



4×4 Cube Stack & Octahedron Net

45° Parallel Sunlight Projection

Q.No & Type	Question Stem & Options
P01.Q01 NAT (+4 / 0)	Refer to Fig P1.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **3** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P01.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **7 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P01.Q03 NAT (+4 / 0)	Refer to Fig P1.2. A square sheet of paper of side **9 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **1.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P01.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **5** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?
P01.Q05 NAT (+4 / 0)	Refer to Fig P1.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **4** full rotations, how many full rotations will Gear S undergo?
P01.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **9 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P01.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P01.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P01.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q.

Q.No & Type	Question Stem & Options
	(B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P01.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P01.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P01.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full. (C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P01.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P01.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P01.Q15 MCQ (+3 / −0.5)	Refer to Fig P1.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P01.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Bengali (বাংলা)** typeface: *"বুলবুল পাখি ময়না টিয়ে"*. Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) modulated 45-degree brush stress, triangular loop in 'ব', and full matra crossbar (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems

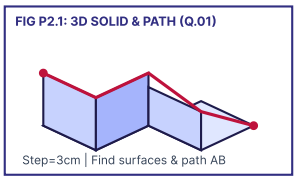
Q.No & Type	Question Stem & Options
P01.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length *L*, fixed at points 1 and 2 separated by distance *d* < *2L* on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P01.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (De Stijl, Impressionism, Pop Art, Baroque, Art Nouveau): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) B (De Stijl [Piet Mondrian grid], Impressionism [Renoir brushwork], Pop Art [Lichtenstein comic dots], Baroque [Vermeer chiaroscuro], Art Nouveau [Mucha Jugend floral curves]) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P01.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Perforated Wooden Peacock Artifact (Jali Woodcarving)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) D (Finishing, Sanding, Drilling, Chiselling — powder coating and sheet-metal punching do not apply to solid carved wood) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P01.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth
P01.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P01.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Sanchi Stupa (MP), Humayun's Tomb (Delhi), Sun Temple Konark (Odisha), Meenakshi Temple (Madurai), Brihadeeswarar Temple (Thanjavur), Shore Temple (Mahabalipuram). (A) Humayun's Tomb (28.59°N) → Sanchi (23.48°N) → Konark (19.89°N) → Mahabalipuram (12.62°N) → Thanjavur (10.78°N) → Madurai (9.92°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P01.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P01.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**?

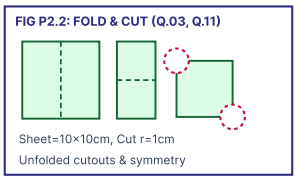
Q.No & Type	Question Stem & Options
	(A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P01.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P01.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P01.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference (2πr) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius 2πr (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P01.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo' (C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P01.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P01.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P01.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification

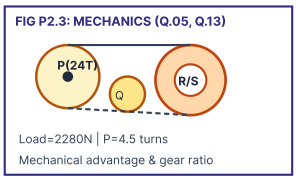
Q.No & Type	Question Stem & Options
P01.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P01.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P01.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P01.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

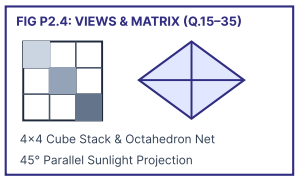
Practice Paper 02: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)









Q.No & Type	Question Stem & Options
P02.Q01 NAT (+4 / 0)	Refer to Fig P2.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **2** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P02.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **8 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P02.Q03 NAT (+4 / 0)	Refer to Fig P2.2. A square sheet of paper of side **10 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **2.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P02.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **6** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?

Q.No & Type	Question Stem & Options
P02.Q05 NAT (+4 / 0)	Refer to Fig P2.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **5** full rotations, how many full rotations will Gear S undergo?
P02.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **10 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P02.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P02.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P02.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q. (B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P02.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P02.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P02.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full.

Q.No & Type	Question Stem & Options
	(C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P02.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P02.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P02.Q15 MCQ (+3 / −0.5)	Refer to Fig P2.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P02.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Gujarati (ગુજરાતી)** typeface: *"નંદલાલ કલા મંદિર"*. Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) absence of top shirorekha, open counter in 'ધ', and bulbous terminal in 'દ' (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems
P02.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length L, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P02.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Renaissance, Baroque, Arts and Crafts, Art Deco, Bauhaus): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) C (Renaissance [1400–1600] → Baroque [1600–1750] → Arts and Crafts [1880–1920] → Bauhaus [1919–1933] → Art Deco [1920–1940]) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P02.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Channapatna Lacquered Wooden Rattle (Ivory-wood Lathe Craft)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) B (Sizing, Turning, Drilling, Lacquering — natural vegetable dye lac sticks are pressed against spinning wood on a lathe) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P02.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth

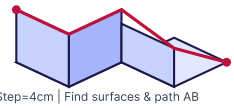
Q.No & Type	Question Stem & Options
P02.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P02.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Martand Sun Temple (J&K), Hawa Mahal (Jaipur), Rani ki Vav (Patan), Ajanta Caves (Maharashtra), Gol Gumbaz (Bijapur), Padmanabhaswamy Temple (Trivandrum). (A) Martand (33.76°N) → Hawa Mahal (26.92°N) → Rani ki Vav (23.85°N) → Ajanta (20.55°N) → Gol Gumbaz (16.83°N) → Trivandrum (8.48°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P02.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P02.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**? (A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P02.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P02.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P02.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference ($2\pi r$) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius $2\pi r$ (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P02.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo'

Q.No & Type	Question Stem & Options
	(C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P02.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P02.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P02.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification
P02.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P02.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P02.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P02.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

Practice Paper 03: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

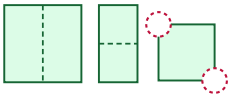
Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)

FIG P3.1: 3D SOLID & PATH (Q.01)



Step=4cm | Find surfaces & path AB

FIG P3.2: FOLD & CUT (Q.03, Q.11)



Sheet=11×11cm, Cut r=2cm

Unfolded cutouts & symmetry

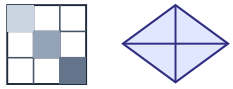
FIG P3.3: MECHANICS (Q.05, Q.13)



Load=2520N | P=5.0 turns

Mechanical advantage & gear ratio

FIG P3.4: VIEWS & MATRIX (Q.15–35)



4×4 Cube Stack & Octahedron Net

45° Parallel Sunlight Projection

Q.No & Type	Question Stem & Options
P03.Q01 NAT (+4 / 0)	Refer to Fig P3.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **3** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P03.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **9 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P03.Q03 NAT (+4 / 0)	Refer to Fig P3.2. A square sheet of paper of side **11 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **1.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P03.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **7** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?
P03.Q05 NAT (+4 / 0)	Refer to Fig P3.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **6** full rotations, how many full rotations will Gear S undergo?
P03.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **11 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P03.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P03.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P03.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q.

Q.No & Type	Question Stem & Options
	(B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P03.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
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P03.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P03.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P03.Q15 MCQ (+3 / −0.5)	Refer to Fig P3.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P03.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Gurmukhi (ਗੁਰਮੁਖੀ)** typeface: "*ਗਰਮ ਅਨਾਰ ਬਾਗ਼ਾ*". Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) heavy geometric shirorekha, squared-off loop in 'ਗ', and uniform monolinear stem (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems

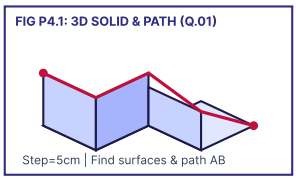
Q.No & Type	Question Stem & Options
P03.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length *L*, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P03.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Cubism, Surrealism, Dadaism, Abstract Expressionism, Minimalism): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) A (Analytic Cubism [Picasso faceted planes], Dada [Duchamp readymade], Surrealism [Dali dreamscape], Abstract Expressionism [Pollock action drip], Minimalism [Judd geometric module]) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P03.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Bastar Dhokra Brass Bull Figurine (Lost-Wax Non-Ferrous Casting)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) C (Clay Core Making, Bee-Wax Thread Coiling, Clay Mould Luting, Molten Brass Pouring — runner channel connects tail and leg) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P03.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth
P03.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P03.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Kangra Fort (HP), Taj Mahal (Agra), Khajuraho (MP),Karla Caves (Lonavala), Hampi Vitthala Temple (Karnataka), Mattancherry Palace (Kochi). (A) Kangra (32.09°N) → Taj Mahal (27.17°N) → Khajuraho (24.85°N) → Karla (18.78°N) → Hampi (15.33°N) → Kochi (9.96°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P03.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P03.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**?

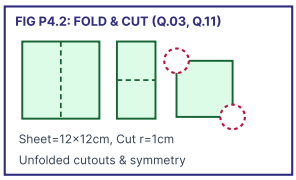
Q.No & Type	Question Stem & Options
	(A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P03.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P03.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P03.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference (2πr) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius 2πr (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P03.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo' (C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P03.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P03.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P03.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification

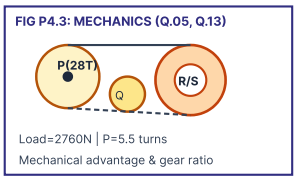
Q.No & Type	Question Stem & Options
P03.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P03.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P03.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P03.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

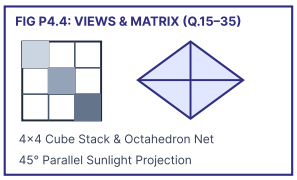
Practice Paper 04: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)









Q.No & Type	Question Stem & Options
P04.Q01 NAT (+4 / 0)	Refer to Fig P4.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **2** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P04.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **10 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P04.Q03 NAT (+4 / 0)	Refer to Fig P4.2. A square sheet of paper of side **12 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **2.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P04.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **8** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?

Q.No & Type	Question Stem & Options
P04.Q05 NAT (+4 / 0)	Refer to Fig P4.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **7** full rotations, how many full rotations will Gear S undergo?
P04.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **12 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P04.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P04.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P04.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q. (B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P04.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P04.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P04.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full.

Q.No & Type	Question Stem & Options
	(C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P04.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P04.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P04.Q15 MCQ (+3 / −0.5)	Refer to Fig P4.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P04.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Tamil (தமிழ்)** typeface: *"நட்டாங்கல்லு கலை"*. Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) high contrast between horizontal hairline and vertical stem, stepped serif in 'க' (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems
P04.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length L, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P04.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Byzantine, Gothic, Rococo, Neoclassicism, Romanticism): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) D (Byzantine mosaic → Gothic pointed vault → Rococo pastel ornament → Neoclassicism severe symmetry → Romanticism sublime storm) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P04.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Moradabad Engraved Brass Surahi (Sheet Spinning & Nakashi)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) A (Sheet Circle Cutting, Lathe Spinning/Forming, Brazing/Soldering, Chisel Engraving & Polishing) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P04.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth

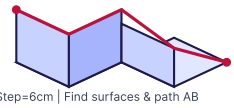
Q.No & Type	Question Stem & Options
P04.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P04.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Golden Temple (Amritsar), Qutub Minar (Delhi), Modhera Sun Temple (Gujarat), Bibi Ka Maqbara (Aurangabad), Charminar (Hyderabad), Rock Fort (Trichy). (A) Amritsar (31.62°N) → Delhi (28.52°N) → Modhera (23.58°N) → Aurangabad (19.90°N) → Hyderabad (17.36°N) → Trichy (10.83°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P04.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P04.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**? (A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P04.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P04.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P04.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference (2πr) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius 2πr (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P04.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo'

Q.No & Type	Question Stem & Options
	(C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P04.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P04.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P04.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification
P04.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P04.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P04.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P04.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

Practice Paper 05: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

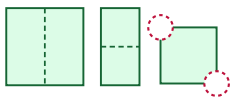
Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)

FIG P5.1: 3D SOLID & PATH (Q.01)



Step=6cm | Find surfaces & path AB

FIG P5.2: FOLD & CUT (Q.03, Q.11)



Sheet=13×13cm, Cut r=2cm

Unfolded cutouts & symmetry

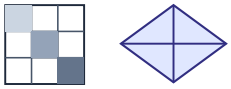
FIG P5.3: MECHANICS (Q.05, Q.13)



Load=3000N | P=6.0 turns

Mechanical advantage & gear ratio

FIG P5.4: VIEWS & MATRIX (Q.15–35)



4×4 Cube Stack & Octahedron Net

45° Parallel Sunlight Projection

Q.No & Type	Question Stem & Options
P05.Q01 NAT (+4 / 0)	Refer to Fig P5.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **3** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P05.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **11 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P05.Q03 NAT (+4 / 0)	Refer to Fig P5.2. A square sheet of paper of side **13 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **1.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P05.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **9** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?
P05.Q05 NAT (+4 / 0)	Refer to Fig P5.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **8** full rotations, how many full rotations will Gear S undergo?
P05.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **13 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P05.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P05.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P05.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q.

Q.No & Type	Question Stem & Options
	(B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P05.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P05.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P05.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full. (C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P05.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P05.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P05.Q15 MCQ (+3 / −0.5)	Refer to Fig P5.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P05.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Devanagari (देवनागरी)** typeface: "*अडगुल डिज़ाइन केंद्र*". Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) humanist sans-serif modulation, open knot in 'ग', and sheared terminal in 'ल' (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems
P05.Q17 MCQ (+3 /	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length L, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at

Q.No & Type	Question Stem & Options
−0.5)	all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P05.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Ukiyo-e Woodblock, Impressionism, Post-Impressionism, Fauvism, Expressionism): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) B (Hokusai flat outlines → Monet light study → Van Gogh swirling impasto → Matisse non-naturalistic colour → Munch psychological distortion) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P05.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Bidriware Silver-Inlaid Black Metal Vase (Bidar, Karnataka)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) D (Zinc-Copper Alloy Casting, File Smoothing, Chisel Groove Etching, Pure Silver Wire Inlay & Ammonium Chloride Soil Blackening) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P05.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth
P05.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P05.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Leh Palace (Ladakh), Fatehpur Sikri (UP), Udaigiri Caves (Vidisha), Elephanta Caves (Mumbai), Belur Chennakesava (Karnataka), Vivekananda Rock (Kanyakumari). (A) Leh (34.16°N) → Fatehpur Sikri (27.09°N) → Vidisha (23.53°N) → Mumbai (18.96°N) → Belur (13.16°N) → Kanyakumari (8.08°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P05.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P05.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**? (A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)

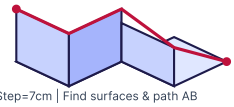
Q.No & Type	Question Stem & Options
P05.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P05.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P05.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference (2πr) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius 2πr (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P05.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo' (C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P05.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P05.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P05.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification
P05.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)

Q.No & Type	Question Stem & Options
P05.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P05.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P05.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

Practice Paper 06: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

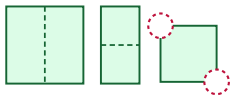
Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)

FIG P6.1: 3D SOLID & PATH (Q.01)



Step=7cm | Find surfaces & path AB

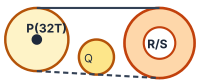
FIG P6.2: FOLD & CUT (Q.03, Q.11)



Sheet=14×14cm, Cut r=1cm

Unfolded cutouts & symmetry

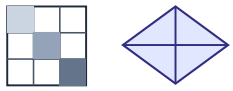
FIG P6.3: MECHANICS (Q.05, Q.13)



Load=3240N | P=6.5 turns

Mechanical advantage & gear ratio

FIG P6.4: VIEWS & MATRIX (Q.15–35)



4×4 Cube Stack & Octahedron Net

45° Parallel Sunlight Projection

Q.No & Type	Question Stem & Options
P06.Q01 NAT (+4 / 0)	Refer to Fig P6.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **2** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P06.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **12 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P06.Q03 NAT (+4 / 0)	Refer to Fig P6.2. A square sheet of paper of side **14 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **2.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P06.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **4** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?
P06.Q05 NAT (+4 / 0)	Refer to Fig P6.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **9** full rotations, how many full rotations will Gear S undergo?

Q.No & Type	Question Stem & Options
P06.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **14 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P06.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P06.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P06.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q. (B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P06.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P06.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P06.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full. (C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P06.Q13 MSQ (+4 /	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s)

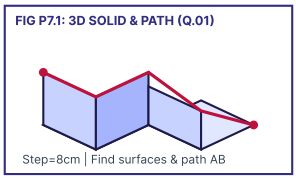
Q.No & Type	Question Stem & Options
−1)	is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P06.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P06.Q15 MCQ (+3 / −0.5)	Refer to Fig P6.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P06.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Kannada (ಕನ್ನಡ)** typeface: "*ಕರಾವಳಿ ಶಿಲ್ಪಕಲೆ*". Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) rounded bowl geometry, hooked headstroke (talakattu), and uniform circular counters (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems
P06.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length L, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P06.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Constructivism, Futurism, Suprematism, Memphis Milano, Streamline Moderne): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) C (Malevich black square → Balla dynamic motion lines → Rodchenko diagonal agitation → Loewy aerodynamic teardrop → Sottsass postmodern laminate) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P06.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Blue Pottery Glazed Tile (Jaipur Quartz-Frit Craft)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) B (Quartz Powder Dough Kneading, Mould Pressing, Oxide Brush Painting, Low-Fire Kiln Glazing — contains zero clay) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P06.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth
P06.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls?

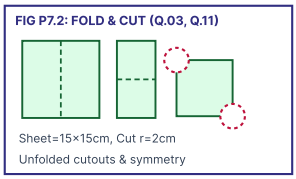
Q.No & Type	Question Stem & Options
	(A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P06.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Pinjore Gardens (Haryana), Chittorgarh Fort (Rajasthan), Champaner-Pavagadh (Gujarat), Ellora Kailasa (Maharashtra), Badami Caves (Karnataka), Rameswaram Temple (TN). (A) Pinjore (30.79°N) → Chittorgarh (24.88°N) → Champaner (22.48°N) → Ellora (20.02°N) → Badami (15.92°N) → Rameswaram (9.28°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P06.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P06.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**? (A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P06.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P06.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P06.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference (2πr) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius 2πr (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P06.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo' (C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth

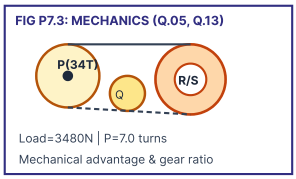
Q.No & Type	Question Stem & Options
P06.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P06.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P06.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification
P06.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P06.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P06.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P06.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

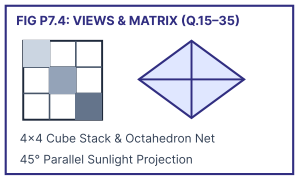
Practice Paper 07: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)









Q.No & Type	Question Stem & Options
P07.Q01 NAT (+4 / 0)	Refer to Fig P7.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **3** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P07.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **13 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P07.Q03 NAT (+4 / 0)	Refer to Fig P7.2. A square sheet of paper of side **15 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **1.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P07.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **5** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?
P07.Q05 NAT (+4 / 0)	Refer to Fig P7.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **10** full rotations, how many full rotations will Gear S undergo?
P07.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **15 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P07.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P07.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P07.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q. (B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.

Q.No & Type	Question Stem & Options
P07.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P07.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P07.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full. (C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P07.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P07.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P07.Q15 MCQ (+3 / −0.5)	Refer to Fig P7.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P07.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Malayalam (മലയാളം)** typeface: *"കളരിപ്പയറ്റ് കേന്ദ്രം"*. Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) looped arch proportions, wide horizontal aperture, and modulated curve stress (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems
P07.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length L, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve

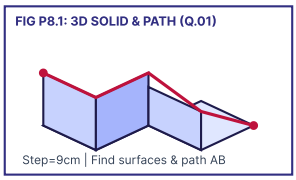
Q.No & Type	Question Stem & Options
P07.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Mysore Painting, Warli, Madhubani, Kalamkari, Pattachitra): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) B (Mysore gesso gold-leaf → Warli white rice-paste triangles → Madhubani double-line kachni → Kalamkari natural pen-resist → Pattachitra bold mineral outlines) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P07.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Aranmula Kannadi Metal Mirror (Kerala Bell-Metal Craft)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) C (Copper-Tin Alloy Lost-Wax Casting, Crucible Firing, Multi-Stage Emery Lapping, Velvet Oil Buffing) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P07.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth
P07.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P07.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Badrinath Temple (Uttarakhand), Bara Imambara (Lucknow), Mahabodhi Temple (Bodh Gaya), Lingaraja Temple (Bhubaneswar), Lepakshi (AP), Suchindram Temple (TN). (A) Badrinath (30.74°N) → Lucknow (26.87°N) → Bodh Gaya (24.69°N) → Bhubaneswar (20.24°N) → Lepakshi (13.80°N) → Suchindram (8.15°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P07.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P07.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**? (A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P07.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the

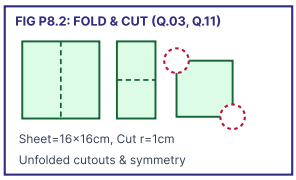
Q.No & Type	Question Stem & Options
	bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P07.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P07.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference (2πr) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius 2πr (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P07.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo' (C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P07.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P07.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P07.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification
P07.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P07.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°

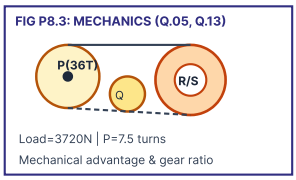
Q.No & Type	Question Stem & Options
P07.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P07.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

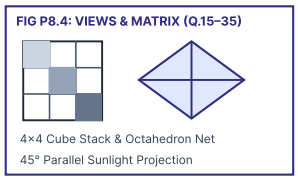
Practice Paper 08: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)









Q.No & Type	Question Stem & Options
P08.Q01 NAT (+4 / 0)	Refer to Fig P8.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **2** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P08.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **14 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P08.Q03 NAT (+4 / 0)	Refer to Fig P8.2. A square sheet of paper of side **16 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **2.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P08.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **6** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?
P08.Q05 NAT (+4 / 0)	Refer to Fig P8.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **11** full rotations, how many full rotations will Gear S undergo?
P08.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **16 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P08.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**?

Q.No & Type	Question Stem & Options
	(A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P08.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P08.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q. (B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P08.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P08.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P08.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full. (C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P08.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.

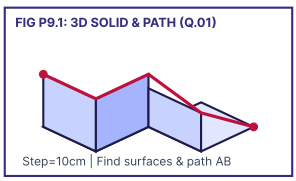
Q.No & Type	Question Stem & Options
P08.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P08.Q15 MCQ (+3 / −0.5)	Refer to Fig P8.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P08.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Odia (ଓଡ଼ିଆ)** typeface: *"କୋଣାର୍କ ସୂର୍ଯ୍ୟ ମନ୍ଦିର"*. Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) distinctive circular umbrella canopy (chhatra) above characters and crisp vertical stems (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems
P08.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length L, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P08.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Gandhara Sculpture, Chola Bronze, Mughal Miniature, Bengal School, Progressive Artists' Group): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) A (Greco-Buddhist drapery → Lost-wax Nataraja → Perso-Indian court folio → Abanindranath wash technique → Souza/Raza modernist oil) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P08.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Kondapalli Wooden Toy (Tella Poniki Softwood Craft)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) A (Seasoning Softwood, Component Carving, Tamarind-Seed Paste Luting [Makku], Vegetable/Watercolour Painting) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P08.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth
P08.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences

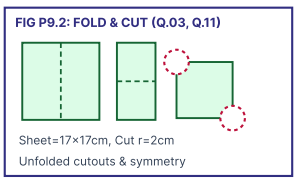
Q.No & Type	Question Stem & Options
P08.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Hidimba Devi Temple (Manali), Amber Fort (Jaipur), Mandu Jahaz Mahal (MP), Daulatabad Fort (Maharashtra), Halebidu (Karnataka), Bekal Fort (Kerala). (A) Manali (32.25°N) → Jaipur (26.98°N) → Mandu (22.36°N) → Daulatabad (19.94°N) → Halebidu (13.21°N) → Bekal (12.39°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P08.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P08.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**? (A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P08.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P08.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P08.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference ($2\pi r$) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius $2\pi r$ (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P08.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo' (C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P08.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis

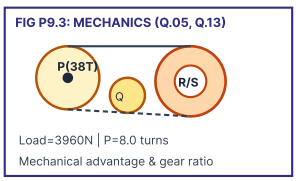
Q.No & Type	Question Stem & Options
P08.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P08.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification
P08.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P08.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P08.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P08.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

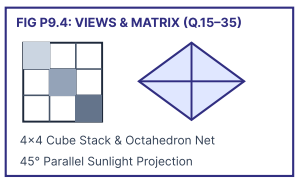
Practice Paper 09: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)









Q.No & Type	Question Stem & Options
P09.Q01 NAT (+4 / 0)	Refer to Fig P9.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **3** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height

Q.No & Type	Question Stem & Options
	1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P09.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **15 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P09.Q03 NAT (+4 / 0)	Refer to Fig P9.2. A square sheet of paper of side **17 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **1.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P09.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **7** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?
P09.Q05 NAT (+4 / 0)	Refer to Fig P9.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **12** full rotations, how many full rotations will Gear S undergo?
P09.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **17 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P09.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section. (C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P09.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P09.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q. (B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P09.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y.

Q.No & Type	Question Stem & Options
	(C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P09.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P09.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full. (C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P09.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P09.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers. (B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P09.Q15 MCQ (+3 / −0.5)	Refer to Fig P9.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P09.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Telugu (తెలుగు)** typeface: "కూచిపూడి నాట్యము". Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) tick-shaped headstroke (talakattu), fluid circular bowls, and high baseline bounce (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems
P09.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length L, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P09.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Art Nouveau, Vienna Secession, Plakatstil, Swiss International Style, Postmodern Punk Typography): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) D (Whiplash botanical curves → Klimt gold geometry → Hohlwein flat poster object → Muller-Brockmann asymmetric Akzidenz grid → Weingart/Carson deconstructed type)

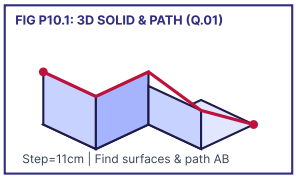
Q.No & Type	Question Stem & Options
	(C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P09.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **Thanjavur Bobblehead Dancing Doll (Terracotta/Papier-Mache)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) D (Two-Part Clay Moulding, Sun Drying, Counterweight Base Balancing, Gimbal Wire Hook Suspension) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P09.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth
P09.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P09.Q22 MCQ (+3 / −0.5)	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Shankaracharya Temple (Srinagar), Jantar Mantar (Delhi), Victoria Memorial (Kolkata), Ramappa Temple (Telangana), Gingee Fort (TN), Krishnapuram Palace (Kerala). (A) Srinagar (34.08°N) → Delhi (28.62°N) → Kolkata (22.54°N) → Ramappa (18.26°N) → Gingee (12.25°N) → Krishnapuram (9.15°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P09.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P09.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**? (A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P09.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder

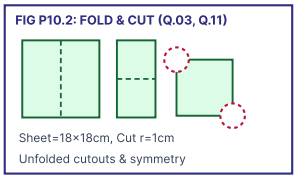
Q.No & Type	Question Stem & Options
P09.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P09.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference (2πr) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius 2πr (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P09.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo' (C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P09.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P09.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height (B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P09.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification
P09.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P09.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P09.Q34 MCQ (+3 /	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing

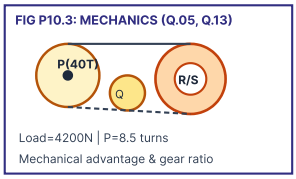
Q.No & Type	Question Stem & Options
−0.5)	suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P09.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

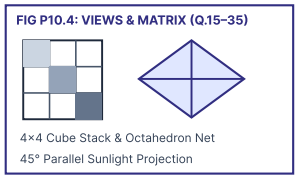
Practice Paper 10: 35-Question Timed Daily Sprint (119 Marks — 40 Minutes)

Section I (Q.01–Q.06): NAT (+4m, 0 Neg) | **Section II (Q.07–Q.14):** MSQ (+4/+3/+2/+1 Partial, −1 Neg) | **Section III (Q.15–Q.35):** MCQ (+3m, −0.5 Neg)









Q.No & Type	Question Stem & Options
P10.Q01 NAT (+4 / 0)	Refer to Fig P10.1. A cylindrical base (diameter 8 cm, height 1 cm) has a central vertical square post (side 2 cm, total height 6 cm). **2** square hollow blocks (outer side 5 cm, inner square hole 2 cm × 2 cm, height 1 cm each) are stacked concentrically around the post, with a 0.5 cm step offset between rings. How many total outer visible & bottom surfaces does the resulting stacked assembly have?
P10.Q02 NAT (+4 / 0)	A waffle cone of inner radius **3 cm** and vertical height **16 cm** is completely filled with ice-cream and topped with a perfect hemisphere of radius **3 cm**. If the density of ice-cream is **0.9 g/cm³** and π = 3.14, calculate the total mass of the ice-cream in grams (up to two decimal places).
P10.Q03 NAT (+4 / 0)	Refer to Fig P10.2. A square sheet of paper of side **18 cm** is folded in half vertically and then folded in half horizontally to form a smaller square. Circular quadrant cuts of radius **2.0 cm** are made at three corners (the folded center corner, one two-fold edge corner, and the four-layer open corner). When unfolded completely, what is the total area (in cm²) of all cut-out pieces removed from the original sheet? (Use π = 3.14)
P10.Q04 NAT (+4 / 0)	Equal-sized red and yellow unit cubes are stacked to form a solid **4 × 4 × 4** cube (64 unit cubes total). The ratio of the total number of yellow cubes to red cubes in the entire structure is **1 : 3**. If an isometric camera view reveals exactly **8** yellow cubes on the three visible outer faces, how many yellow cubes are **NOT** visible in that view?
P10.Q05 NAT (+4 / 0)	Refer to Fig P10.3. Pulley P (diameter 40 cm) drives Pulley Q (diameter 10 cm) via a non-slip belt. Gear R (20 teeth) is fixed coaxially to Pulley Q and meshes with Gear S (40 teeth). If Pulley P undergoes **13** full rotations, how many full rotations will Gear S undergo?
P10.Q06 NAT (+4 / 0)	A monospaced font has an exact character width of **0.5 cm** (including the space character). A single-line headline in a poster contains **18 words**, where each word consists of exactly **5 letters** separated by single spaces (with no leading or trailing spaces). What is the exact total length of the text line in cm?
P10.Q07 MSQ (+4 / −1)	A regular hexagonal prism (height 10 cm, base side 6 cm), a cube (side 10 cm), and a right circular cylinder (height 10 cm, diameter 10 cm) are intersected by single flat cutting planes at various angles. Which of the following statements is/are **TRUE**? (A) The cylinder can reveal an elliptical, circular, or rectangular cross-section, including a square. (B) Both the cube and the hexagonal prism can reveal a regular hexagonal cross-section.

Q.No & Type	Question Stem & Options
	(C) Both the cube and the hexagonal prism can reveal an equilateral triangle cross-section. (D) The cylinder can reveal a triangular cross-section when cut diagonally through the base center.
P10.Q08 MSQ (+4 / −1)	Four symmetrical 3D green blocks (A, B, C, D) are placed inside cylindrical containers and molten wax is poured up to their horizontal mid-plane (marked by a yellow dashed line). Assuming wax does not stick to the blocks, which block(s) will leave an exact, undamaged negative impression when lifted straight up vertically? (A) Block A: A cube with outward-tapering frustum bosses whose widest diameter is at the bottom base. (B) Block B: Two horizontal cylinders intersecting at 90° where wax is poured exactly up to their central axes. (C) Block C: An inverted frustum pyramid (wider at the top than at the bottom) with 10° outward draft on all side walls. (D) Block D: A stepped block with an internal dovetail slot below the wax line.
P10.Q09 MSQ (+4 / −1)	A garden secateurs (plant pruner) has two handles pivoted at a central bolt and two studs P and Q on the inner handle faces. Which spring configuration(s) fitted between P and Q will successfully return the blades to the **OPEN** position after squeezing? (A) A helical compression coil spring mounted over telescopic guide pins between P and Q. (B) A volute (conical ribbon) compression spring seated on studs P and Q. (C) A helical tension spring with end hooks looped around P and Q that pulls P and Q together. (D) A double-loop torsion spring (hairpin/safety-pin style) anchored on the studs.
P10.Q10 MSQ (+4 / −1)	In a clinical study of 300 patients across two viral variants V1 and V2, Drug X achieved 45/50 successes on V1 (90%) and 160/250 on V2 (64%), giving 205/300 total. Drug Y achieved 220/250 on V1 (88%) and 30/50 on V2 (60%), giving 250/300 total. Which statement(s) is/are **TRUE** (Simpson's Paradox)? (A) If the variant is known to be V1, Drug X has a higher probability of success than Drug Y. (B) If the variant is known to be V2, Drug X also has a higher probability of success than Drug Y. (C) Because 250/300 > 205/300, Drug Y is superior to Drug X for both individual variants. (D) If a clinic must stock only one treatment when both variants are equally prevalent, Drug X is superior.
P10.Q11 MSQ (+4 / −1)	A square tile of 6 × 6 grid units has two triangular tabs protruding from its top and right edges and two matching triangular notches cut out of its bottom and left edges. Which transformation(s) allow this tile to form a gapless, overlap-free planar tessellation? (A) Pure horizontal and vertical translation without rotation. (B) 90° clockwise rotation of alternating tiles. (C) Reflection across the main diagonal followed by translation. (D) 180° rotation when tabs and notches are centrally symmetric.
P10.Q12 MSQ (+4 / −1)	Four household liquid dispensers are examined: (1) a flip-top shampoo bottle, (2) a squeeze-valve sports water bottle, (3) a trigger-spray glass cleaner, and (4) a push-pump liquid soap bottle. Which statement(s) is/are **TRUE**? (A) At least two models rely on one-way check valves to function. (B) Models (3) and (4) can dispense liquid without being lifted or tilted when 1/4 full. (C) At least one model uses a class-1 or class-2 lever mechanism. (D) All four models can dispense liquid without being lifted when 1/8 full.
P10.Q13 MSQ (+4 / −1)	An animation of four planets (Red, Green, Blue, Brown) orbiting a central star shows angular velocities ω_Red = 4 rad/s, ω_Green = 2 rad/s, ω_Blue = 1 rad/s, and ω_Brown = 2/3 rad/s. Which statement(s) is/are **TRUE**? (A) When Planet Blue completes 2 orbits, Planet Green completes 4 orbits. (B) When Planet Blue completes 3 orbits, Planet Brown completes 2 orbits. (C) When Planet Green completes 1 orbit, Planet Red completes 2 orbits. (D) When Planet Red completes 6 orbits, Planet Brown completes 1 orbit.
P10.Q14 MSQ (+4 / −1)	A food-delivery platform is redesigning its mobile app ecosystem for Tier-2 Indian cities. Which stakeholder group(s) should the UX research team interview during the contextual inquiry phase? (A) Delivery partners navigating narrow lanes on two-wheelers.

Q.No & Type	Question Stem & Options
	(B) Restaurant kitchen staff managing order-preparation tablets. (C) Customer-support agents handling refund and delay escalations. (D) Existing and lapsed end-consumers ordering meals on the app.
P10.Q15 MCQ (+3 / −0.5)	Refer to Fig P10.4. The Top View of a stack of unit sugar cubes is a complete 4 × 4 grid (all 16 base squares occupied). The Front View and Right Side View are identical stepped profiles with column heights (4, 4, 3, 2) from left to right. What is the **MINIMUM** number of unit cubes required to build this stack? (A) 22 (B) 25 (C) 28 (D) 31
P10.Q16 MCQ (+3 / −0.5)	Four words are printed in a specific **Assamese (অসমীয়া)** typeface: "*কাজিৰঙা জাতীয় উদ্যান*". Which typographic feature is the primary diagnostic clue for selecting the matching target word from the four options? (A) Uniform monolinear dot-matrix pixelation (B) diagonal cross-stroke in 'ৰ', acute triangular apex, and classical serif terminals (C) Reverse-contrast heavy horizontals and thin verticals (D) Disconnected stencil bridges on all vertical stems
P10.Q17 MCQ (+3 / −0.5)	A red dot is attached to the midpoint of two non-elastic strings P and Q of equal length L, fixed at points 1 and 2 separated by distance $d < 2L$ on a flat plane. If at least one string is kept fully stretched (taut) at all times, what is the shape of the closed boundary path traced by the red dot? (A) Figure-eight lemniscate (B) Smooth ellipse (C) Vesica piscis (lens formed by two intersecting circular arcs) (D) Hyperbolic X-curve
P10.Q18 MCQ (+3 / −0.5)	Select the correct sequence that names the five art/design movements shown sequentially from left to right (Brutalism, Metabolism, High-Tech Architecture, Deconstructivism, Parametricism): (A) Pop Art, Baroque, De Stijl, Impressionism, Art Nouveau (B) A (Le Corbusier Chandigarh beton brut → Kurokawa plug-in capsule → Rogers/Piano exposed service ducts → Gehry titanium fragmentation → Hadid continuous NURBS shell) (C) Impressionism, De Stijl, Baroque, Art Nouveau, Pop Art (D) De Stijl, Art Nouveau, Pop Art, Impressionism, Baroque
P10.Q19 MCQ (+3 / −0.5)	A traditional Indian craft artifact — **cane & Split-Bamboo Mora Stool (Northeast India Basketry)** — is produced using heritage tools and processes. Which option lists the relevant operations involved in its making (not necessarily in sequence)? (A) Injection Moulding, Ultrasonic Welding, Anodizing (B) B (Culm Splitting, Strip Width Sizing, Hyperboloid Criss-Cross Weaving, Rim Lashing & Smoking) (C) Extrusion, Powder Coating, Laser Sintering (D) Drop Forging, Galvanizing, Broaching
P10.Q20 MCQ (+3 / −0.5)	A public transit bus seat is being ergonomically designed for the adult Indian population. Which anthropometric specification ensures that **MOST** passengers can sit comfortably with their feet resting flat on the floor and without hip compression? (A) Seat height = 95th percentile sitting height; Seat breadth = 5th percentile hip breadth (B) Seat height = 5th/50th percentile popliteal sitting height; Seat breadth = 95th percentile hip breadth (C) Seat height = 95th percentile popliteal height; Seat breadth = 50th percentile hip breadth (D) Seat height = 5th percentile sitting height; Seat breadth = 5th percentile hip breadth
P10.Q21 MCQ (+3 / −0.5)	Four mobile checkout wireframes (Screens A, B, C, D) display: (1) Delivery Address (mutually exclusive: Home vs. Work), (2) Packaging Option (mutually exclusive: Cardboard, Foam, Hard Case), and (3) Delivery Instructions (multiple allowed: Do not ring bell, Leave at guard). Which screen **INCORRECTLY** uses standard UI controls? (A) Screen A: Radio buttons for Address & Packaging; Checkboxes for Delivery Instructions (B) Screen B: Dropdown for Address; Radio buttons for Packaging; Text field for Instructions (C) Screen C: Radio buttons where BOTH 'Yes' and 'No' are simultaneously selected for Share Live Location (D) Screen D: Single-select radio buttons for Packaging and multi-select checkboxes for Preferences
P10.Q22 MCQ (+3 /	Arrange the following six Indian architectural monuments in their correct geographical order from **North to South**: Tawang Monastery (Arunachal), Kamakhya Temple (Guwahati), Nalanda Mahavihara (Bihar),

Q.No & Type	Question Stem & Options
−0.5)	Jagannath Temple (Puri), Amaravati Stupa (AP), Gangaikonda Cholapuram (TN). (A) Tawang (27.58°N) → Guwahati (26.16°N) → Nalanda (25.13°N) → Puri (19.80°N) → Amaravati (16.57°N) → Gangaikonda (11.20°N) (B) Reverse South-to-North order (C) West-to-East longitudinal order (D) Chronological dynasty order
P10.Q23 MCQ (+3 / −0.5)	Four visual diagrams illustrate Gestalt principles from left to right: (1) a 6×6 grid of circles grouped by blue vs. magenta columns, (2) two intersecting sinusoidal dotted waves of alternating colours, (3) three columns of triangles separated by wide vertical alleys, and (4) an incomplete 5-pointed star with open vertices. Identify the sequence: (A) Closure, Proximity, Continuity, Similarity (B) Similarity, Continuity, Proximity, Closure (C) Continuity, Similarity, Closure, Proximity (D) Proximity, Closure, Similarity, Continuity
P10.Q24 MCQ (+3 / −0.5)	A solid copper block with an equilateral triangular base has a constant 100°C heat source applied at vertex P. Points A, B, C, and D lie along the opposite top ridge, where a semi-circular groove is machined between P and C. Neglecting heat losses, which point reaches 100°C **EARLIEST**? (A) Point A (shortest unobstructed straight-line conduction path from P) (B) Point B (C) Point C (obstructed by air gap of groove) (D) Point D (longest path)
P10.Q25 MCQ (+3 / −0.5)	A hollow vertical open cylinder (height = diameter) is illuminated by parallel sunlight at 45° from the top-left. When viewed in isometric perspective, what is the shape of the lit boundary cast on the inner concave wall of the cylinder? (A) A flat horizontal straight line (B) A crescent-like curve starting at the top rim and dipping to the bottom-right inner wall (C) A vertical rectangular stripe (D) Complete darkness inside the cylinder
P10.Q26 MCQ (+3 / −0.5)	A circular disc of diameter *D* (parallel to the ground) and a solid sphere of diameter *D* are floating above flat ground under parallel sunlight at 45°. When viewed from directly above (Top View), what are the shapes of their shadows on the ground? (A) Disc shadow = Circle; Sphere shadow = Ellipse (B) Disc shadow = Ellipse; Sphere shadow = Ellipse (C) Disc shadow = Circle; Sphere shadow = Circle (D) Disc shadow = Ellipse; Sphere shadow = Circle
P10.Q27 MCQ (+3 / −0.5)	A rigid rod PQ whose length equals the circumference (2πr) of a fixed green circle rests tangentially with P at the top of the circle. The rod rolls without slipping around the circle until end Q touches P. What is the locus traced by point Q? (A) A circle of radius 2πr (B) An involute spiral of decreasing radius curling inward to P (C) A straight vertical line (D) A symmetric semicircle of radius πr
P10.Q28 MCQ (+3 / −0.5)	When the phrase **"See you Bye"** is spoken by an animated character, which sequence of 5 mouth visemes (lip shapes) is correct? (A) Wide teeth 'S/ee' → Tightly rounded 'oo' → Closed bilabial line 'B' → Wide open 'ah' → Slightly parted 'ee' (B) Closed lips → Wide teeth → Rounded 'oo' → Closed lips → Rounded 'oo' (C) Open mouth throughout all 5 frames (D) Rounded 'oo' → Closed lips → Wide teeth → Closed lips → Wide teeth
P10.Q29 MCQ (+3 / −0.5)	A 3D pipe object is rotated **90° clockwise about the Y-axis**, then **180° anti-clockwise about the X-axis**, followed by **90° anti-clockwise about the Y-axis** (all viewed from the positive axis toward the origin). This composite 3-step rotation is equivalent to a single: (A) 180° rotation about the Z-axis (B) 90° clockwise rotation about the X-axis (C) Identity transformation (no change) (D) 90° anti-clockwise rotation about the Z-axis
P10.Q30 MCQ (+3 / −0.5)	In a domestic kitchen designed for a single user, three activities are performed: (Task 1) washing utensils in a deep sink, (Task 2) rolling chapatis, and (Task 3) chopping vegetables. For optimal spinal and elbow ergonomics, how should the counter heights be arranged? (A) All three tasks at the exact same standard counter height

Q.No & Type	Question Stem & Options
	(B) Tasks 1, 2, and 3 at three different ergonomic heights (Sink rim highest, chopping medium, heavy rolling slightly lower) (C) Tasks 1 and 2 at the same height (D) Sink rim lowest at knee level
P10.Q31 MCQ (+3 / −0.5)	A monument is photographed from four distances using four lenses of focal lengths 24mm, 50mm, 85mm, and 200mm such that the foreground subject remains the same size in the frame. How does increasing the focal length affect the background? (A) Higher focal length magnifies the background and compresses perceived depth (B) Higher focal length shrinks the background and widens the field of view (C) Focal length has no effect on background scale (D) 24mm gives maximum background magnification
P10.Q32 MCQ (+3 / −0.5)	Which material among the following engineered and natural woods will produce the **LIGHTEST** piece of furniture for the exact same volumetric design dimensions? (A) Medium Density Fibreboard (MDF, ~750 kg/m³) (B) Indian Teak Wood (~650 kg/m³) (C) Particle Board / Chipboard (~550 kg/m³) (D) High-Density Marine Plywood (~700 kg/m³)
P10.Q33 MCQ (+3 / −0.5)	Two layered watch dials P (bottom) and Q (top, with a 90° sector window) start at 12:00. Dial Q rotates 90° clockwise every 15 minutes; Dial P rotates 45° anti-clockwise every 30 minutes. At 13:30 hrs (90 minutes later), what are their net angular positions relative to 12:00? (A) Q is at +180° (upside down); P is at −135° anti-clockwise (B) Q is at 0°; P is at −90° (C) Q is at +90°; P is at +135° (D) Both return to 0°
P10.Q34 MCQ (+3 / −0.5)	Biomimicry pairs: Match the biological organism (P: Kingfisher beak, Q: Humpback whale flipper tubercles, R: Gecko toe-pad setae, S: Burdock seed hooks) with the product it inspired (1: Climbing suction pads, 2: Velcro hook-and-loop fastener, 3: Shinkansen bullet-train nose cone, 4: Low-drag wind-turbine blades): (A) P-3, Q-4, R-1, S-2 (B) P-3, Q-1, R-4, S-2 (C) P-4, Q-2, R-1, S-3 (D) P-1, Q-3, R-2, S-4
P10.Q35 MCQ (+3 / −0.5)	An animator is drawing a fish jumping out of water along a parabolic arc (Point A: launch from water, Point B: mid-rise, Point C: highest apex of jump, Point D: re-entry splash). To convey realistic gravity (slow-out on rise, slow-in on fall), where must the animator draw the **MAXIMUM** spatial density of frames? (A) At Point A (B) At Point B (C) At Point C (the apex, where vertical velocity approaches zero) (D) At Point D

Section II: Complete 88-Day Part A Daily 35-Question Schedule (Oct 21, 2026 – Jan 16, 2027)

Follow this day-by-day 35-question schedule every morning or evening alongside your 2-hour Part B studio session. It systematically interleaves the **10 Original 35-Question Practice Papers** in Section I with the **10 Official Past CEED Papers (2017–2026)** and targeted topic-twin drills.

Day & Date	Assigned 35-Question Daily Part A Sprint (35–45 Mins)	Core Cognitive Focus	Target & Log
Day 01 Wed, 21 Oct 2026 *(Day 0 (Launch))*	**Official CEED 2017 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2017 spatial & analytical items	78+ / 119 Score: ____
Day 02 Thu, 22 Oct 2026 *(Day 01 of 87)*	**Official CEED 2018 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2018 spatial & analytical items	78+ / 119 Score: ____

Day & Date	Assigned 35-Question Daily Part A Sprint (35–45 Mins)	Core Cognitive Focus	Target & Log
Day 03 Fri, 23 Oct 2026 *(Day 02 of 87)*	**Official CEED 2019 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2019 spatial & analytical items	78+ / 119 Score: ____
Day 04 Sat, 24 Oct 2026 *(Day 03 of 87)*	**Official CEED 2020 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2020 spatial & analytical items	78+ / 119 Score: ____
Day 05 Sun, 25 Oct 2026 *(Day 04 of 87)*	**Official CEED 2021 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2021 spatial & analytical items	78+ / 119 Score: ____
Day 06 Mon, 26 Oct 2026 *(Day 05 of 87)*	**Official CEED 2022 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2022 spatial & analytical items	78+ / 119 Score: ____
Day 07 Tue, 27 Oct 2026 *(Day 06 of 87)*	**Official CEED 2023 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2023 spatial & analytical items	78+ / 119 Score: ____
Day 08 Wed, 28 Oct 2026 *(Day 07 of 87)*	**Official CEED 2024 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2024 spatial & analytical items	78+ / 119 Score: ____
Day 09 Thu, 29 Oct 2026 *(Day 08 of 87)*	**Official CEED 2025 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2025 spatial & analytical items	78+ / 119 Score: ____
Day 10 Fri, 30 Oct 2026 *(Day 09 of 87)*	**Official CEED 2026 Part A Core Sprint:** Solve Q.01 to Q.35 (NAT + MSQ + First 17 MCQs)	Baseline diagnostic on CEED 2026 spatial & analytical items	78+ / 119 Score: ____
Day 11 Sat, 31 Oct 2026 *(Day 10 of 87)*	**Original Practice Paper 01 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 12 Sun, 01 Nov 2026 *(Day 11 of 87)*	**Original Practice Paper 02 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 13 Mon, 02 Nov 2026 *(Day 12 of 87)*	**Original Practice Paper 03 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 14 Tue, 03 Nov 2026 *(Day 13 of 87)*	**Original Practice Paper 04 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 15 Wed, 04 Nov 2026 *(Day 14 of 87)*	**Original Practice Paper 05 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 16 Thu, 05 Nov 2026 *(Day 15 of 87)*	**Original Practice Paper 06 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 17 Fri, 06 Nov 2026 *(Day 16 of 87)*	**Original Practice Paper 07 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____

Day & Date	Assigned 35-Question Daily Part A Sprint (35–45 Mins)	Core Cognitive Focus	Target & Log
Day 18 Sat, 07 Nov 2026 *(Day 17 of 87)*	**Original Practice Paper 08 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 19 Sun, 08 Nov 2026 *(Day 18 of 87)*	**Original Practice Paper 09 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 20 Mon, 09 Nov 2026 *(Day 19 of 87)*	**Original Practice Paper 10 (Section I of this Workbook):** Full 35Q Sprint (Q.01–Q.35)	2027-pattern 6 NAT + 8 Partial-MSQ + 21 MCQ timed drill	82+ / 119 Score: ____
Day 21 Tue, 10 Nov 2026 *(Day 20 of 87)*	**Hybrid 35Q Sprint:** CEED 2017 (Q.19–Q.44: 26 MCQs) + CEED 2018 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 22 Wed, 11 Nov 2026 *(Day 21 of 87)*	**Hybrid 35Q Sprint:** CEED 2018 (Q.19–Q.44: 26 MCQs) + CEED 2019 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 23 Thu, 12 Nov 2026 *(Day 22 of 87)*	**Hybrid 35Q Sprint:** CEED 2019 (Q.19–Q.44: 26 MCQs) + CEED 2020 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 24 Fri, 13 Nov 2026 *(Day 23 of 87)*	**Hybrid 35Q Sprint:** CEED 2020 (Q.19–Q.44: 26 MCQs) + CEED 2021 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 25 Sat, 14 Nov 2026 *(Day 24 of 87)*	**Hybrid 35Q Sprint:** CEED 2021 (Q.19–Q.44: 26 MCQs) + CEED 2022 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 26 Sun, 15 Nov 2026 *(Day 25 of 87)*	**Hybrid 35Q Sprint:** CEED 2022 (Q.19–Q.44: 26 MCQs) + CEED 2023 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 27 Mon, 16 Nov 2026 *(Day 26 of 87)*	**Hybrid 35Q Sprint:** CEED 2023 (Q.19–Q.44: 26 MCQs) + CEED 2024 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 28 Tue, 17 Nov 2026 *(Day 27 of 87)*	**Hybrid 35Q Sprint:** CEED 2024 (Q.19–Q.44: 26 MCQs) + CEED 2025 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 29 Wed, 18 Nov 2026 *(Day 28 of 87)*	**Hybrid 35Q Sprint:** CEED 2025 (Q.19–Q.44: 26 MCQs) + CEED 2026 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 30 Thu, 19 Nov 2026 *(Day 29 of 87)*	**Hybrid 35Q Sprint:** CEED 2026 (Q.19–Q.44: 26 MCQs) + CEED 2017 (Q.01–Q.09: 8 NAT + 1 MSQ)	High-speed MCQ elimination + zero-error NAT verification	85+ / 119 Score: ____
Day 31 Fri, 20 Nov 2026 *(Day 30 of 87)*	**Topic-Intensive 35Q Sprint #01:** 3D Surface & Edge Counting Master Sprint (All 2017–2026 NATs + Original Q01s)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 32 Sat, 21 Nov 2026 *(Day 31 of 87)*	**Topic-Intensive 35Q Sprint #02:** Paper Folding, Punching, Shibori Dyeing & Unfolded Nets (35-Question Visual Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____

Day & Date	Assigned 35-Question Daily Part A Sprint (35–45 Mins)	Core Cognitive Focus	Target & Log
Day 33 Sun, 22 Nov 2026 *(Day 32 of 87)*	**Topic-Intensive 35Q Sprint #03:** Mechanical Linkages, Gears, Pulleys, Cams & Springs (35-Question Physics Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 34 Mon, 23 Nov 2026 *(Day 33 of 87)*	**Topic-Intensive 35Q Sprint #04:** Orthographic Views, Minimum Cube Stacks & Perspective Angles (35-Question Spatial Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 35 Tue, 24 Nov 2026 *(Day 34 of 87)*	**Topic-Intensive 35Q Sprint #05:** 45-Degree Parallel Sunlight Shadow Projections & Optics (35-Question Ray-Tracing Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 36 Wed, 25 Nov 2026 *(Day 35 of 87)*	**Topic-Intensive 35Q Sprint #06:** Typography, Font Anatomy & Indic Scripts (Bengali, Tamil, Gurmukhi, Gujarati, Devanagari)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 37 Thu, 26 Nov 2026 *(Day 36 of 87)*	**Topic-Intensive 35Q Sprint #07:** Animation Principles, Keyframe Spacing, Walk-Cycles & Viseme Lip-Sync (35-Question Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 38 Fri, 27 Nov 2026 *(Day 37 of 87)*	**Topic-Intensive 35Q Sprint #08:** Seamless Tessellations, Tangrams, Stamp Impressions & Matrix Overlaps (35-Question Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 39 Sat, 28 Nov 2026 *(Day 38 of 87)*	**Topic-Intensive 35Q Sprint #09:** Indian Monuments Geography, Art Movements & Traditional Wood/Metal Crafts (35Q Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 40 Sun, 29 Nov 2026 *(Day 39 of 87)*	**Topic-Intensive 35Q Sprint #10:** Anthropometry (5th/50th/95th Percentiles), UI/UX Components & Fitts's Law (35Q Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 41 Mon, 30 Nov 2026 *(Day 40 of 87)*	**Topic-Intensive 35Q Sprint #11:** 3D Surface & Edge Counting Master Sprint (All 2017–2026 NATs + Original Q01s)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 42 Tue, 01 Dec 2026 *(Day 41 of 87)*	**Topic-Intensive 35Q Sprint #12:** Paper Folding, Punching, Shibori Dyeing & Unfolded Nets (35-Question Visual Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 43 Wed, 02 Dec 2026 *(Day 42 of 87)*	**Topic-Intensive 35Q Sprint #13:** Mechanical Linkages, Gears, Pulleys, Cams & Springs (35-Question Physics Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 44 Thu, 03 Dec 2026 *(Day 43 of 87)*	**Topic-Intensive 35Q Sprint #14:** Orthographic Views, Minimum Cube Stacks & Perspective Angles (35-Question Spatial Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 45 Fri, 04 Dec 2026 *(Day 44 of 87)*	**Topic-Intensive 35Q Sprint #15:** 45-Degree Parallel Sunlight Shadow Projections & Optics (35-Question Ray-Tracing Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 46 Sat, 05 Dec 2026 *(Day 45 of 87)*	**Topic-Intensive 35Q Sprint #16:** Typography, Font Anatomy & Indic Scripts (Bengali, Tamil, Gurmukhi, Gujarati, Devanagari)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 47 Sun, 06 Dec 2026 *(Day 46 of 87)*	**Topic-Intensive 35Q Sprint #17:** Animation Principles, Keyframe Spacing, Walk-Cycles & Viseme Lip-Sync (35-Question Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____

Day & Date	Assigned 35-Question Daily Part A Sprint (35–45 Mins)	Core Cognitive Focus	Target & Log
Day 48 Mon, 07 Dec 2026 *(Day 47 of 87)*	**Topic-Intensive 35Q Sprint #18:** Seamless Tessellations, Tangrams, Stamp Impressions & Matrix Overlaps (35-Question Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 49 Tue, 08 Dec 2026 *(Day 48 of 87)*	**Topic-Intensive 35Q Sprint #19:** Indian Monuments Geography, Art Movements & Traditional Wood/Metal Crafts (35Q Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 50 Wed, 09 Dec 2026 *(Day 49 of 87)*	**Topic-Intensive 35Q Sprint #20:** Anthropometry (5th/50th/95th Percentiles), UI/UX Components & Fitts's Law (35Q Sprint)	Pattern recognition across question twins (see Trend Report Table 2.1)	88+ / 119 Score: ____
Day 51 Thu, 10 Dec 2026 *(Day 50 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2017):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 52 Fri, 11 Dec 2026 *(Day 51 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2018):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 53 Sat, 12 Dec 2026 *(Day 52 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2019):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 54 Sun, 13 Dec 2026 *(Day 53 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2020):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 55 Mon, 14 Dec 2026 *(Day 54 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2021):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 56 Tue, 15 Dec 2026 *(Day 55 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2022):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 57 Wed, 16 Dec 2026 *(Day 56 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2023):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 58 Thu, 17 Dec 2026 *(Day 57 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2024):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 59 Fri, 18 Dec 2026 *(Day 58 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2025):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 60 Sat, 19 Dec 2026 *(Day 59 of 87)*	**Speed-Capped Past Paper Sprint (CEED 2026):** 35 Selected High-Difficulty Qs in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 61 Sun, 20 Dec 2026 *(Day 60 of 87)*	**Speed-Capped Original Paper 01:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 62 Mon, 21 Dec 2026 *(Day 61 of 87)*	**Speed-Capped Original Paper 02:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____

Day & Date	Assigned 35-Question Daily Part A Sprint (35–45 Mins)	Core Cognitive Focus	Target & Log
Day 63 Tue, 22 Dec 2026 *(Day 62 of 87)*	**Speed-Capped Original Paper 03:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 64 Wed, 23 Dec 2026 *(Day 63 of 87)*	**Speed-Capped Original Paper 04:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 65 Thu, 24 Dec 2026 *(Day 64 of 87)*	**Speed-Capped Original Paper 05:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 66 Fri, 25 Dec 2026 *(Day 65 of 87)*	**Speed-Capped Original Paper 06:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 67 Sat, 26 Dec 2026 *(Day 66 of 87)*	**Speed-Capped Original Paper 07:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 68 Sun, 27 Dec 2026 *(Day 67 of 87)*	**Speed-Capped Original Paper 08:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 69 Mon, 28 Dec 2026 *(Day 68 of 87)*	**Speed-Capped Original Paper 09:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 70 Tue, 29 Dec 2026 *(Day 69 of 87)*	**Speed-Capped Original Paper 10:** Re-solve all 35 Qs with Dimensional Variants in 35 Mins	Compression from 45m down to 35m (1.0 min/question pace)	92+ / 119 Score: ____
Day 71 Wed, 30 Dec 2026 *(Day 70 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #01:** 15 Error-Log Traps + 20 Qs from CEED 2017 / Original Set 01	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 72 Thu, 31 Dec 2026 *(Day 71 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #02:** 15 Error-Log Traps + 20 Qs from CEED 2018 / Original Set 02	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 73 Fri, 01 Jan 2027 *(Day 72 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #03:** 15 Error-Log Traps + 20 Qs from CEED 2019 / Original Set 03	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 74 Sat, 02 Jan 2027 *(Day 73 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #04:** 15 Error-Log Traps + 20 Qs from CEED 2020 / Original Set 04	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 75 Sun, 03 Jan 2027 *(Day 74 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #05:** 15 Error-Log Traps + 20 Qs from CEED 2021 / Original Set 05	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 76 Mon, 04 Jan 2027 *(Day 75 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #06:** 15 Error-Log Traps + 20 Qs from CEED 2022 / Original Set 06	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 77 Tue, 05 Jan 2027 *(Day 76 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #07:** 15 Error-Log Traps + 20 Qs from CEED 2023 / Original Set 07	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____

Day & Date	Assigned 35-Question Daily Part A Sprint (35–45 Mins)	Core Cognitive Focus	Target & Log
Day 78 Wed, 06 Jan 2027 *(Day 77 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #08:** 15 Error-Log Traps + 20 Qs from CEED 2024 / Original Set 08	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 79 Thu, 07 Jan 2027 *(Day 78 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #09:** 15 Error-Log Traps + 20 Qs from CEED 2025 / Original Set 09	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 80 Fri, 08 Jan 2027 *(Day 79 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #10:** 15 Error-Log Traps + 20 Qs from CEED 2026 / Original Set 10	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 81 Sat, 09 Jan 2027 *(Day 80 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #11:** 15 Error-Log Traps + 20 Qs from CEED 2017 / Original Set 01	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 82 Sun, 10 Jan 2027 *(Day 81 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #12:** 15 Error-Log Traps + 20 Qs from CEED 2018 / Original Set 02	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 83 Mon, 11 Jan 2027 *(Day 82 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #13:** 15 Error-Log Traps + 20 Qs from CEED 2019 / Original Set 03	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 84 Tue, 12 Jan 2027 *(Day 83 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #14:** 15 Error-Log Traps + 20 Qs from CEED 2020 / Original Set 04	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 85 Wed, 13 Jan 2027 *(Day 84 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #15:** 15 Error-Log Traps + 20 Qs from CEED 2021 / Original Set 05	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 86 Thu, 14 Jan 2027 *(Day 85 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #16:** 15 Error-Log Traps + 20 Qs from CEED 2022 / Original Set 06	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 87 Fri, 15 Jan 2027 *(Day 86 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #17:** 15 Error-Log Traps + 20 Qs from CEED 2023 / Original Set 07	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____
Day 88 Sat, 16 Jan 2027 *(Day 87 of 87)*	**Peak Dress-Rehearsal 35Q Sprint #18:** 15 Error-Log Traps + 20 Qs from CEED 2024 / Original Set 08	MSQ partial-credit optimization (+4/+3/+2) & zero negative bleed	96+ / 119 Score: ____

Section III: Master Answer Key & Rapid Formula Compendium

Use the master key below to score your 10 Original 35-Question Practice Papers (out of 119 marks per paper) and review the core mathematical and spatial formulas tested in CEED Part A.

Practice Set	Section I: NAT Key (Q01–Q06)	Section II: MSQ Key (Q07–Q14)	Section III: MCQ Key (Q15–Q35)
Paper 01 (119 Marks)	**Q01:** 31 **Q02:** 110.21 g **Q03:** 9.42 cm^2 **Q04:** 11	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:**

Practice Set	Section I: NAT Key (Q01–Q06)	Section II: MSQ Key (Q07–Q14)	Section III: MCQ Key (Q15–Q35)
	Q05: 8 **Q06:** 26.5 cm	**Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C
Paper 02 (119 Marks)	**Q01:** 23 **Q02:** 118.69 g **Q03:** 37.68 cm^2 **Q04:** 10 **Q05:** 10 **Q06:** 29.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C
Paper 03 (119 Marks)	**Q01:** 31 **Q02:** 127.17 g **Q03:** 9.42 cm^2 **Q04:** 9 **Q05:** 12 **Q06:** 32.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C
Paper 04 (119 Marks)	**Q01:** 23 **Q02:** 135.65 g **Q03:** 37.68 cm^2 **Q04:** 8 **Q05:** 14 **Q06:** 35.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C
Paper 05 (119 Marks)	**Q01:** 31 **Q02:** 144.13 g **Q03:** 9.42 cm^2 **Q04:** 7 **Q05:** 16 **Q06:** 38.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C
Paper 06 (119 Marks)	**Q01:** 23 **Q02:** 152.6 g **Q03:** 37.68 cm^2 **Q04:** 12 **Q05:** 18 **Q06:** 41.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C
Paper 07 (119 Marks)	**Q01:** 31 **Q02:** 161.08 g **Q03:** 9.42 cm^2 **Q04:** 11 **Q05:** 20 **Q06:** 44.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C

Practice Set	Section I: NAT Key (Q01–Q06)	Section II: MSQ Key (Q07–Q14)	Section III: MCQ Key (Q15–Q35)
Paper 08 (119 Marks)	**Q01:** 23 **Q02:** 169.56 g **Q03:** 37.68 cm^2 **Q04:** 10 **Q05:** 22 **Q06:** 47.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C
Paper 09 (119 Marks)	**Q01:** 31 **Q02:** 178.04 g **Q03:** 9.42 cm^2 **Q04:** 9 **Q05:** 24 **Q06:** 50.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C
Paper 10 (119 Marks)	**Q01:** 23 **Q02:** 186.52 g **Q03:** 37.68 cm^2 **Q04:** 8 **Q05:** 26 **Q06:** 53.5 cm	**Q07:** A, B, C **Q08:** A, B, C **Q09:** A, B, D **Q10:** A, B, D **Q11:** A, D **Q12:** A, B, C **Q13:** A, B, C, D **Q14:** A, B, C, D	**Q15:** B (25) | **Q16:** B | **Q17:** C | **Q18:** B | **Q19:** B | **Q20:** B | **Q21:** C | **Q22:** A | **Q23:** B | **Q24:** A | **Q25:** B | **Q26:** A | **Q27:** B | **Q28:** A | **Q29:** A | **Q30:** B | **Q31:** A | **Q32:** C | **Q33:** A | **Q34:** A | **Q35:** C

High-Speed Mathematical & Spatial Formula Compendium for Part A

- **3D Surface Counting Rule:** Break any complex solid into (a) Top/Horizontal planar faces, (b) Bottom planar faces, (c) Front/Back vertical faces, (d) Left/Right vertical faces, and (e) Curved cylindrical/conical surfaces. Remember that a 360° closed cylinder has 3 surfaces (top, bottom, 1 curved lateral face), whereas a 270° revolved profile (like CEED 2025 Q02) exposes **two additional flat radial cut planes**!
- **Paper Folding Symmetry Multiplier:** Each fold across an axis doubles the layers. A cut made strictly inside the interior of an n-times folded sheet produces 2^n identical holes. However, a semi-circular or triangular cut made along a *single-fold crease* merges into 1 symmetric hole per crease layer when unfolded, and a cut at the *central four-layer folded vertex* produces 1 single central hole!
- **Minimum Cubes in a Stack (Top + Front + Side Views):** If the Top View is a full $N \times N$ grid (requiring at least 1 cube on every base cell = N^2 cubes) and the Front and Side views have identical column heights $(h_1, h_2, \ldots, h_N)$, place the tall columns along the main diagonal cells (i, i) so each diagonal tower satisfies both the Front and Side height simultaneously! Minimum cubes = $N^2 + \Sigma(h_i - 1)$. For 2026 Q19 with 4×4 base and heights (4,4,3,2), Min = 16 + (3+3+2+1) = **25 cubes**.
- **Gear & Pulley Transmission Law:** For meshing gears or belt-connected pulleys, $N_1 \times D_1 = N_2 \times D_2$ (or $N_1 \times T_1 = N_2 \times T_2$). Coaxial gears/pulleys fixed on the same shaft share the exact same angular velocity (RPM/rotations).
- **Anthropometric Percentile Rule:** Design **reach distances and seat heights** (popliteal height) for the **5th percentile** (or 50th adjustable) so shorter users are not left with dangling feet; design **clearances, doorway heights, and seat hip breadth** for the **95th percentile** so larger users fit without obstruction.